

Delineation Of Groundwater Potential Zones Of Velpula-Vempalli Area

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An integrated survey based on satellite image interpretation corroborated with limited field checks were carried out with a view to delineating the groundwater potential of Velpula-Vempalli area, Kadapa district, Andhra Pradesh falling in Survey of India toposheet no. 57J/07. Under this study six thematic maps-lithological, structural, geomorphological, hydromorphological map, landuse - land cover map, lineament and lineament density map were prepared covering an area of about 3600 km². The lithological and structural map constitutes the geological map. The geological map portrays the area being covered by peninsular gneissic complex comprising mainly granite gneisses and granitoids intruded by basic dolerite dykes and acidic intrusives. The lineaments are associated with late acidic intrusives, fracture filled basic dykes, faults and alignment of stream segments, tanks, topography, etc. The geomorphological map depicts various landforms, such as denudational hill (DH), residual hills (RH), structural hills (SH), structural valley (SV), intermontane valleys (IV), pediment (PD), pediplain (PP) and flood plain. The area has been classified into high potential, moderate potential, low potential and non potential for groundwater development on the basis of hydromorphological studies. Some of the favourable locations have been suggested to impound the excessive runoff so as to augment the groundwater resources of the area.

KEYWORDS

Delineation, Groundwater potential zones, Velpula-Vempalli area, Y.S.R district

1. INTRODUCTION

Groundwater is one of the most valuable natural resources, which supports human health, economic development and ecological diversity. Because of its several inherent qualities, it has become an immensely important and dependable source of water supplies in all climatic regions including both urban and rural areas of developed and developing countries [1]. Water is important source for life and developmental activity and groundwater is the only reliable resource due to dwindling of surface water sources [2]. Providing safe drinking water to rural population is a hectic task for many developing and under developing countries. Since last few decades, millions of handpumps were installed all over India to provide bacteriologically safe drinking water. Groundwater can have some dissolved chemical constituents which may cause chronic health effects [3].

The occurrence and yield potential of groundwater sanctuary are basically controlled by geology, geomorphology and structural set-up of an area. It is important to target groundwater potential zone prior to any planning for groundwater development. Remote sensing has become an indispensable tool for groundwater target-

ing in hard rock areas because the topographic expression and terrain characteristics have direct relation to the geological characteristics of rocks and their structural set-up. Geomorphology exercises a significant control over groundwater regime. An area of high drainage density also increases surface runoff compared to a low drainage density area. Surface water bodies, like river, ponds, etc., can act as recharge zones, enhancing the groundwater potential in the neighborhood [4]. Hence, identification and quantization of these features are important in generating groundwater potential model of a particular area. A geographic information system (GIS) can be used effectively for this purpose to combine different hydrogeological themes objectively and analyse those systematically for demarcating the potential zone [5,6]. The GIS technology provides suitable alternatives for efficient management of large and complex databases. Watershed monitoring and management has been found to be economical and faster with the usage of the capabilities of a GIS [7]. The landform-cum-lineament and lineament density mapping is very useful into delineate groundwater. One of the greatest advantages of using remote sensing data for hydrogeological investigation and monitoring is its ability to generate information in spatial and temporal domain, which is very crucial for successful analysis, prediction and validation [8]. Integration of remote sensing data in GIS environment is very useful in delineating various groundwater potential zone is a meaningful way.

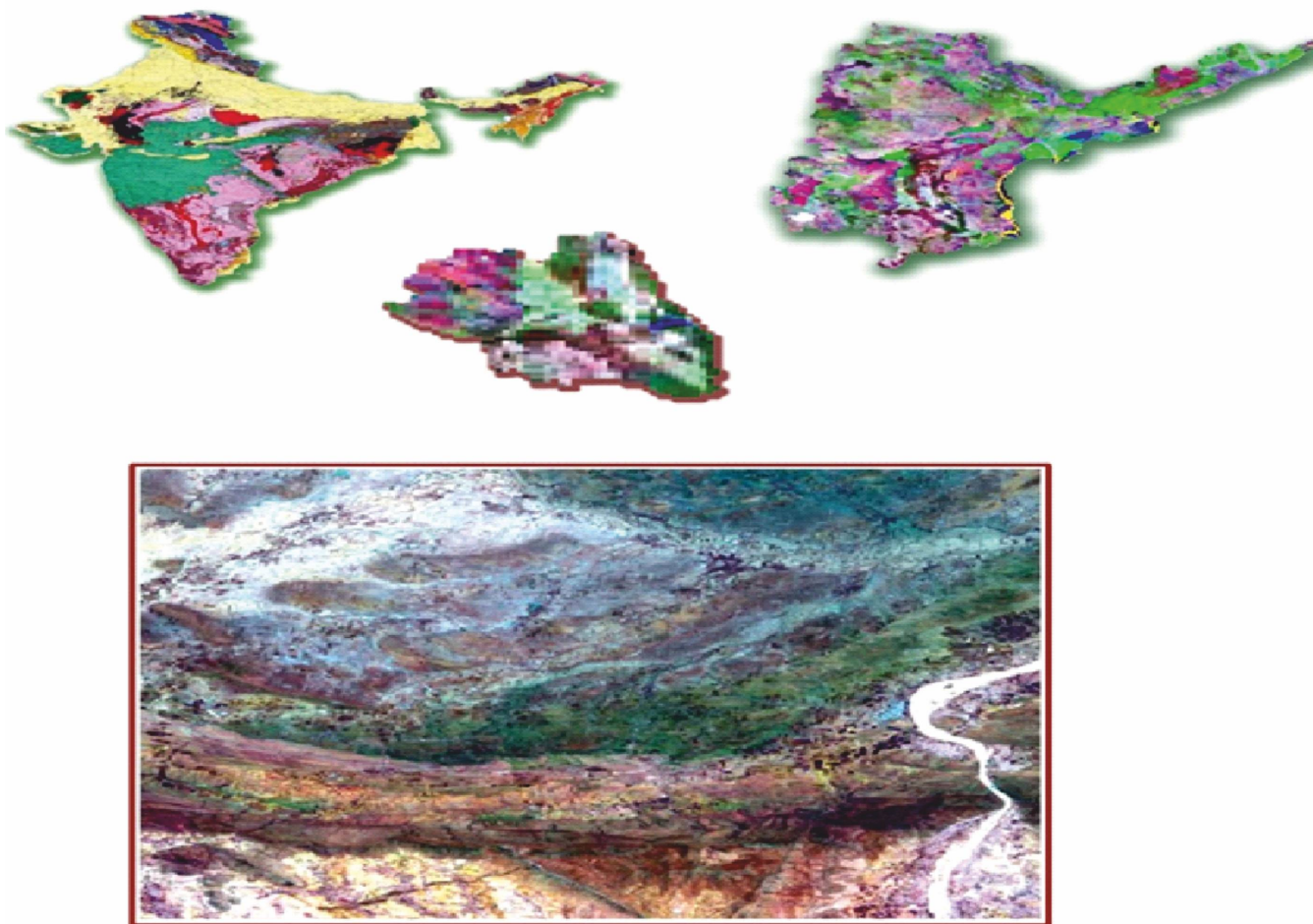


Figure 1. Location map of study area

1.1 Location accessibility

The area of study lies between Velpula-Vempalli bounded by latitude $14^{\circ}30'0''$ N to $14^{\circ}15'0''$ N and longitude $78^{\circ}15'0''$ E to $78^{\circ}30'0''$ E falling in Survey of India toposheet no. 57J/07. The total area covered is approximately 720 km². The state highway from Kadapa to Pulivendula passes in the central part of the area through Vempalli, Vemula and Velpula. The state highway from Vempalli to Lakkireddipalli through Virannagattupalli and Chakrayapeta serve the eastern part of the area. A highway which bifurcates from Kadapa-Pulivendula at Tangadupalli passes through Palagiri and Ulimella serve the western part of the area (Figure 1).

2. METHODOLOGY

Geocoded satellite imagery 57 J/07 (IRS- P6, LISS III) is used to extract the information resource base comprising various litho units, soils, dynamic resource including vegetation cover, forests, surface and subsurface water resource, landuse pattern, drainage and

communication network and habitation distribution. The thematic information pertaining to each resource is prepared by using geocoded satellite imagery and Survey of India toposheet no: 57 J/07. These prefield thematic maps are modified, substantiated to confirm after limited field checks.

At the first stage of survey, standard recognition of elements for geological, geomorphological, hydrological and landuse, land cover maps are made and a tentative map is prepared based on standard image/photograph expressions of selected and entire programme for field check is planned. The purpose of field is to establish the relation between the field characters of different features and their respective image/photographic expression. Dominant recognition element characteristics of particular feature are determined and are made use in extrapolation of features, accurately to a maximum extent with highest degree of accuracy. This includes extrapolation of different features with the help of inter-relation already established between field and image/photo expression of the features. Any de-

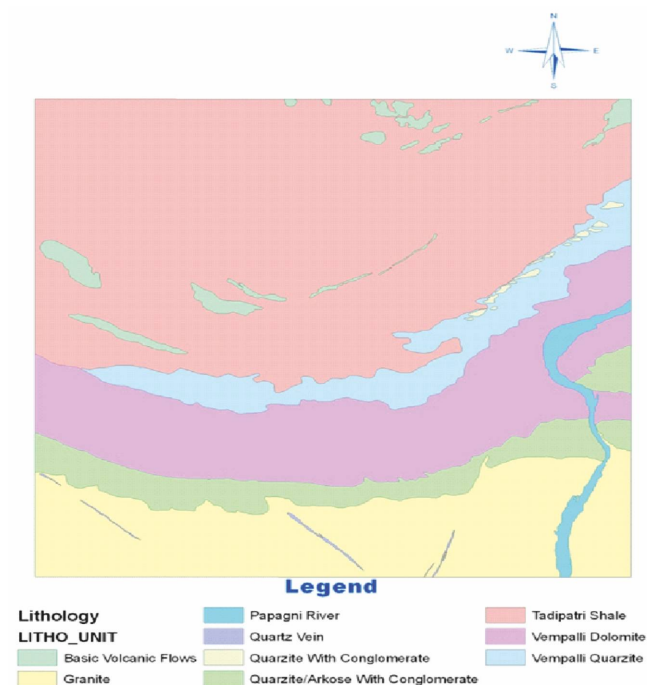


Figure 2. Lithology map (based on image interpretation with limited field checks)

parture from pre-field interpretation observed during field checks is suitably incorporated at this stage and map is finalized after necessary modifications. The central and southern part of the area is hilly terrain when compared to the northern part of the area which is a flat terrain. The central part of the area is occupied by alternate ridges and valleys while the southern part by

Table. 1. Stratigraphic classification of study area

Group	Formation	Lithology
Chitravathi	Tadipatri Pulivendula	Sills shales quartzite
Unconformity		
Papagini mud stone/shale, dolomite	Volcanic flows Vempalli formation Gulcheru	Cherty dolomite quartzite/ conglomerate
Unconformity		
Peninsular Gneissic complex	Granite gneisses, granitoids with acidic and basic intrusive	

high hills with linear ridges and small hillocks. The Papaghni river and its tributaries forms the main drainage in the area. Easterly flowing tributaries namely Idapalapaya vanka and Moganur eru are the two main tributaries to northerly flowing Papaghni river.

2.1 Geology

The study area exposes the rocks belonging to Peninsular Gneissic Complex and Papaghni Nallamalai group of Cuddapah super group of rocks covering an area about 700 km. Peninsular gneissic complex unconformably overlies by a thick sequence of Proterozoic sediments belonging to Papaghni, Chitravathi group Cuddapah supergroup. Lithology map is shown in figure 2. The stratigraphic sequence is given in table 1. The image characteristics of the litho units are given in table 2.

Table 2. Image characteristics of the litho units of study area

Rock unit	Tone	Texture	Drainage pattern	Resistance to erosion	Topography	Vegetation	Soil cover	Human influence
Granite	Lighter	Even	Sub dendritic	Resistant	Pedi plains, pediments	Poor	Thin	Nil
Granite gneiss	Lighter	Coarse	Sub dendritic	Resistant	Incelberg, Residual hills	Medium	Thick	Cultivated confined to low lying channel
Gulcheru quartzite	Medium to dark	Coarse	Sub dendritic	Resistant	Strike ridges	Medium to poor	Absent	Nil
Vempalli formation dolomite	Lighter to medium	Fine to medium	Dendritic parallel	Moderate resistant	Hummocky	Medium to poor	Thin	Cultivated
Cherty dolomite	Lighter to medium	Fine to medium	Dendritic parallel	Moderate resistant	Hummocky	Medium to poor	Thin	Cultivated
Pulivendula quartzite	Lighter to medium	Medium to coarse	Subdendritic, coarse	Resistant	Strike ridges	Good	Absent	Nil
Tadipatri shale	Medium to dark	Fine to medium	Dendritic, dense	Non resistant	Pedi plains cultivated land	Moderate	Thick	Cultivated widely
Dikes and quartz reefs	Lighter to darker	Fine to medium	Dendritic	Moderate	Linear ridge	Moderate	Thin	Nil
Sill bodies	Darker	Fine to medium	Dendritic	Moderate	Linear ridges	Moderate	Thin	Nil

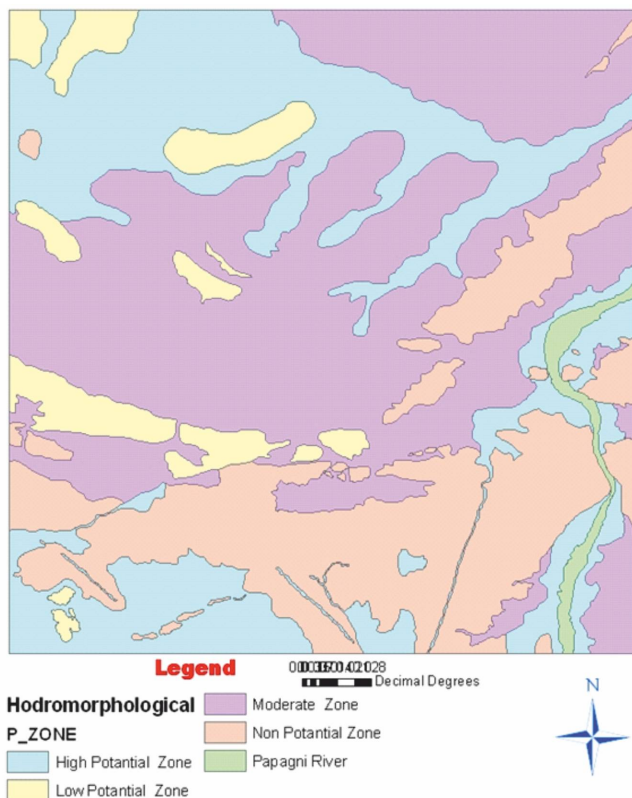


Figure 3. Hydromorphology map of the study area

2.2 Geomorphology

Geomorphology involves study of landforms, reconstruction of process responsible for their origin and study of influence of tectonics in time space frame. The geomorphological mapping includes inventory and classification of landforms. Each landform depend by its composition depth of weathering structural frame and the environment which includes soil cover, hydrology and hydrogeology. The landforms are classified on the basis of mode of origin, relief slope factor and surface cover. The landforms occurring in the area as grouped as denudational hill, residual hill, pediment, pediplain, cuesta, structural hill, structural valley, linear ridge and flood plain. The geomorphologic mapping is important for understanding resource potential, resource utilization, resource depletion and degradation associated with landforms. The geomorphic units developed on various lithostratigraphic units are represented in table 3. Hydromorphology map is shown in figure 3.

2.3 Denudational hill

The Denudational hills are carved out of gneissic and granites occupied in the southern part of area. These landforms are structurally controlled as they are bounded by streams and lineaments. Narrow to wide

Table 3. Hydrogeomorphic units vis-à-vis lithostratigraphy of study area

Map symbol	Geomorphic unit	Lithology	Groundwater prospects
DH	Denudational hill	Sedimentary	Poor
RH	Residual hill	Sedimentary	Poor
PP	Pediplain	Sedimentary outcrops	Poor to mediate
FP	Floodplain	Alluvium (sand, silt, clay)	-
LR	Linear ridge	-	Good to excellent poor

pediments are developed at the basal portion of these hills. Lineaments are expressed as shears, master joint patterns and criss cross patterns. These are observed in the area around Somallapalli north of Vepamanipenta, Eguvapalli, northwest and north of Komarakalva village. The foot slopes covered with boulder to cobble sized blocks of slope debris and weathered to coarse brownish red soils and supports scrubbing and forest vegetation. Mid slopes are covered with huge chunks of slightly modified rock debris. Crest portion at places is slightly weathered and gritty.

2.4 Residual hill

These occurs as isolated low relief hills around the denudational hills as noticed in the southern part of the area around Komarakalva and Chintalamadugupalli. The pediment is narrow around these hills. Residual hills are emplaced by a number of linear basic dyke ridges. These ridges shows a relief of 10 to 30 m. Most of these ridges support scrubby vegetation.

2.5 Tors and corestone plane

Areas of tors and corestone plane mark the dissected pediment and pediplain of the area around Vepamnipenta, Kumarakava, Vemalagondi and Diguvaplli. The tors rise from 3 to 10 m height. corestone planes are also developed in pediplain areas where the basal surface weathering is shallow. Corestone and tors are developed where two sets of vertical jointing and horizontal jointing are more intensive and basal surface of weathering is exposed due to intensive erosion of weathered crust. Soil profile is rather thin in corestone plains and comprising of brownish red sands and coarse gritty sized weathered material. Vegetation usually occupy these rocky waste plains.

2.6 Pediment

The pediment shows 2°-5° slope and associated with

foot slope element of denudational hill, residual hill and ridges as seen in the area around Vepamanipenta to Komarkalva and also around Velpula to Chintalamadugupalli. The pediment is weathered considerably to the depth of few mm to 5 mm. Thin coarse red and brownish sandy soils and skeletal gritty soil covered the pediments due to intensive sheet wash process operating in pediment area. The natural vegetation of the pediment comprises open scrub and grass land at places. The pediments are under dry cultivation and groundwater condition is low to moderate.

2.7 Pediplain

It is gently sloping ramp of 2° - 5° originated by coalescence of several pediment and finally merges with major tributary stream valley and floodplain. Thin to thick weathered crust ranging in thickness from 1 m- 10 m are observed in the pediplain area around Ulimella, Goturu, Muttukuru, Palagiri Chintalajuturu, Gollalaguduru, Alavalapadu, and Veerapunanyinipalli. The ground water prospects in this area is poor to moderate.

2.8 Floodplain

Floodplain form a flat landform with appreciable development of natural levee along the bank of river Papagni. The width of the terrace alluvium ranges from few cm to 5 m. At places thick black cotton soils are occupied in alluvial plains. Alluvium is brownish in colour forming a flat terrain. Floodplain is observed in Tangedipalli, Vempalli, Veerannaguttupalli, Kondavandlapalli and Gondikovuru. The groundwater prospects in this region is good to excellent.

2.9 Structure

Bedding, joint, fracture, lineament and fault are some of the structure elements identified in the area of study. Fractures and lineaments are important structures and the lineaments either coincide with the drainage directions, alignments with the tanks, vegetation or topography.

2.10 Bedding

Bedding is manifested by colour banding or compositional layering and is observed in the formation of Gulcheru quartzite, Vempalli dolomites of Papaghi and Pulivendula quartzite of Chitravathi group. Basement granite does not show bedding. The general trend of bedding in Gulcheru quartzite and Pulivendula quartzite is E-W with shallow dips (5° - 15°) towards north.

3. LINEAMENT

Lineaments are the linear features of tectonic origin

then are identified as long narrow and relatively straight tonal alignments visible in satellite images. A lineament may be a fault, fracture, master joint, a long and linear geological formation, the straight course of streams, vegetation served may be the result of faulting and fracturing and hence it is inferred that they are the areas and zones of increased porosity and permeability in hard rock areas. These have more significance in the ground water studies. Remote sensing data provides useful information to identify structural features and lineaments. The satellite data of IRS-P6 -LISS III imageries of false colour composite have been visually interpreted to identify the lineaments of the area. The data have been checked by field studies and survey of Indian topographical sheets at the 1:50,000 scale. The interpretation of satellite data indicates that the rocks appear to have been deformed repeatedly. The Cuddapah basin is a major synformal structure with minor antiforms and synforms. NE-SW, NNE-SSW and N-S trending faults are prominent in the area. They are developed mainly in Peninsular Gneissic Complex, Gulcheru quartzite and Vempalli formation of Cuddapah super group. The important faults are Rachakuntapalle fault, Eguvapalle fault, Vemula fault, Mabbuchintalapalle fault and Vepamanipenta fault these faults are named after the name of the village. The length of the faults varies from 1.5 to 8.5 km, number of faults are present both in peninsular gneissic complex and overlain the Cuddapah super group of rocks. Most prominent of these are NE-SW extending from basement into Cuddapah supergroup of rocks, Gulcheru and Vempalli formation NNE-SSW faults and north east and south west faults. Three sets of fracture planes are observed in and around Gonayapalli, Vaddipalli, Gollapalli, Eguvapalli. The length of the fractures varies from 1 to 5.5 km. Fracture planes are extending from Peninsular Gneissic Complex-Vempalli dolomite in Gulcheru village with strike line of 5.5 km length. Another fracture plane is observed in the Diguvapalli area. Three sets of lineaments were identified in the area which varies from 3 to 20 km structure map is shown in figure 4. The area around Somalolapalle lineament extends over a strike length of 8 km. Ulimella lineament extends over a strike length of 12 km, with trends NE-SW. The another major lineament extending from Busiredipalle to Gonamakulapalle extending over a strike length of about 11 km. Based on cross-cutting relationship it is established that the NE-SW trending lineaments are the earliest, followed by NW to SE direction and finally NS direction which are acted as vents, for the ground water.

3.1 Lineament density

The lineament present in the study area has varying

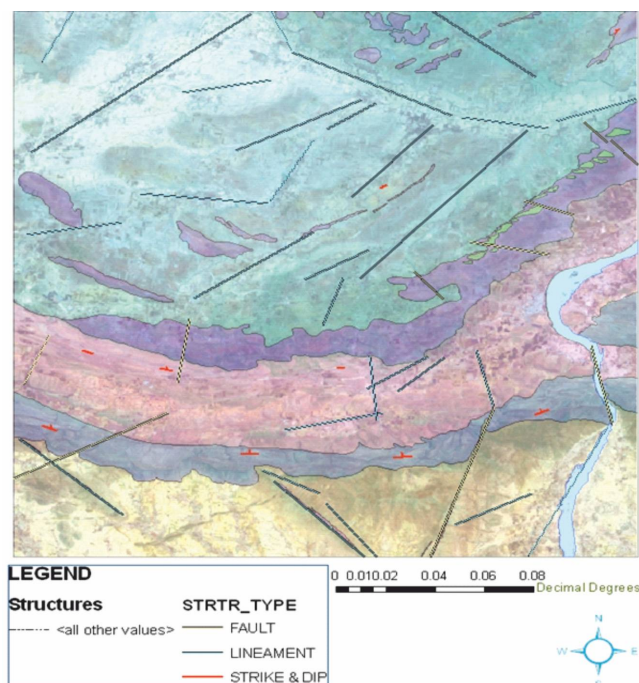


Figure 4. Structure map of the study area

dimensions. Based on the concentration and length of the lineaments, a lineament density map was prepared. The lineament map was superimposed on a grid map of 1 cm² and the total length of lineament passing in each of grid was measured and plotted in the respective grid centers. The values were interconnected by isolines and thus a lineament density map was prepared (Figure 5). Lineaments are the main features that control the occurrence of groundwater. Secondary porosity is imparted by joints and fractures in the areas of higher values of lineament density. The lineament density map reveals the variations of ground water potentiality in the basin. The high lineament density is noticed as isolated patches with small areal extent in the northern and north western parts of the basin indicating high groundwater potential. Medium lineament density patches with limited areal extent indicate a moderate groundwater potential distributed mostly in the central , southern and as a small patches in the western parts of the basin. A large part of the basin area is occupied by low lineament density indicating a poor groundwater potential. Based on the lineament density it is inferred that the groundwater prospects are poor in large part of the study area.

3.2 Hydromorphology

Ground water occurrence in hard rock terrain is confined to certain land form and fractures. As the aquifer material and alluvium is usually confined to certain landform, further lineaments, landform development and

their elevation and distribution is controlled by faults, streams, segments and fractures and structurally exclusive litho units, like dykes and acidic intrusives. It is evident from the above factors that detailed land form mapping cum classification elevation and an understanding of morpho techniques is imperative for ground water exploration in hard rock terrain. The land form in the study area Vemula and Vempalli is grouped under 4 categories as potential land form, moderate potential landform low potential landform and non-potential landform. The basis for classification is degree, density and intensity of the structural disturbances, such as the extended nature of lineaments, land forms and depth of weathering.

3.3 High potential land forms

These include (structurally bound weathered basin) lineament oriented stream valley, pediplain, at the junction of stream and stream valleys under tank reservoir. All lineaments passing through pediplain, dissected pediment and structural valley as seen in the area around Vepamanipeta, Mulakkayalapenta, Komarkalva, Kondavandlapalli, Tangedupalli, Kovvura, Rachakuntapalli, Bumayagaripalli and Busireddipalli.

3.4 Moderately potential landforms

Deeply weathered pediments, stream valleys controlled by structure and impermeable, impervious litho units dissected by lineaments places as seen in the areas Ramireddipalli, Palagiri, Gonumakulapalli, Peddagotur and Chinthajatur.

3.5 Low potential landform

These include hard structural hills dissected by wide space joints around Giddankipalli, northern part of Basireddipalli and eastern part of Veerannagattupalli.

3.6 Non potential landform

Non potential zones are recognized in the imagery based on the absence of intersection of joints, fractures and lineaments. These can be observed in southern part of Bairenkonda formation, eastern part of Malamagalabavi, northern part of Kanumulopalli.

3.7 Landuse land cover

3.7.1 Wetland: Wetland is observed in shale formation which is intersected by streams and also observed along Ulimella, Nallalacheruvu, Muthukuru, Ramireddipalli, Gollaguduru, Pennapadu, Alavapadu, Kothapalli, Kuppalapalli, Chintalamadugupalli, Tallapalli and south of Bumayagaripalli. The wet land nearly constitute 25% of landuse.

3.7.2 Dry land : Dry land is observed in shale which is

not intersected by stream also observed along south of Goturu and north of Gollaguduru, east of Pendluru, Gunduvaripalli and is also observed in granite gneiss formation which is observed in village Komarakalva and Peddannavaripalli.

3.7.3 Barren land : Barren land is also noticed in cherty dolomite formation which is observed in villages in Firangikonda, south of Chintalamadugupalli, east of Idupulapaya vanka, west of Giddangivaripalli, south of Mabbuchintalapalli.

3.7.4 Waste land : Waste land observed towards east of Ammayagaripalli and north of Vempalli formation.

3.7.5 Forest land : Forest land is observed in Gorivikanuma, Gonavaripalli, Idupulapaya, Redlachervu.

4. CONCLUSION AND RECOMMENDATION

Delineation of groundwater potential zones studies were carried out in and around Velpula and Vempalli area, Kadapa district, Andhra Pradesh. Geomorphology mapping helps in understanding resource potential, resource utilization and land degradation. High potential zones are recognized in the imagery along fractures and shear zones, which are observed in Vapamanipenta where the litho unit granite observed with irregular criss-cross fractures and shear zones. Flood plain zones are high potential water body zones which occurs in villages Komarakalva, Kondavandlapalli, Gandikovvur, Tangedupalli, Veerannagattupalli. Moderate potential zones are observed in shale terrains which are seen in Ramireddipalli, Palagiri, Gonumathilapalli, Peddaguturu, Chintalajuturu, Bokkannagaripalli. Low potential zones are observed in cherty dolomite formation which are seen in west of Bumayagaripalli and south of Velamavaripalli. Non potential zones are recognized on the imagery based on the absence of intersection of joints fracture and lineaments and presence of reserve forest at

Gorivikanuma, Ganganapalli.

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